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A student investigated the effect of chewing on the digestion of starch in cooked wheat.

He devised a laboratory model of starch digestion in the human gut. This is the method he used.

1. Volunteers chewed cooked wheat for a set time. The wheat had been cooked in boiling water.
2. This chewed wheat was mixed with water, hydrochloric acid and a protein-digesting enzyme and left at 37 °C for 30 minutes.
3. A buffer was then added to bring the pH to 6.0 and pancreatic amylase was added. This mixture was then left at 37 °C for 120 minutes.
4. Samples of the mixture were removed at 0, 10, 20, 40, 60 and 120 minutes, and the concentration of reducing sugar in each sample was measured.
5. Control experiments were carried out using cooked wheat that had been chopped up in a blender, not chewed.

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What reducing sugar, or sugars, would you expect to be produced during chewing? Give a reason for your answer.

[2 marks]

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In this model of digestion in the human gut, what other enzyme is required for the complete digestion of starch?

[1 mark]

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What was the purpose of step 2, in which samples were mixed with water, hydrochloric acid and pepsin?

[1 mark]

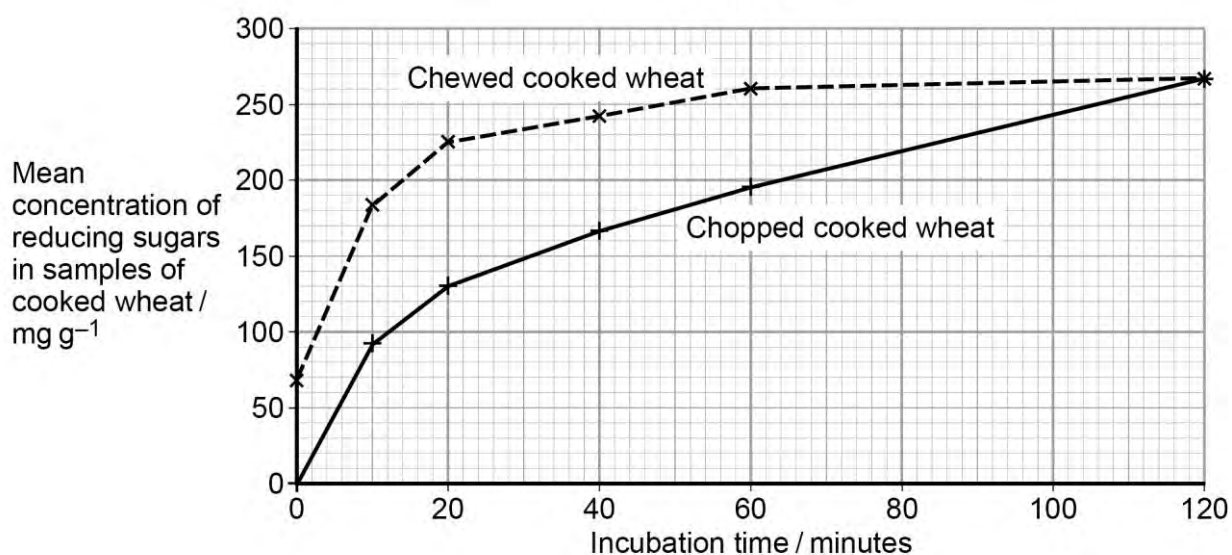
- 0 3 . 4 In the control experiments, cooked wheat was chopped up to copy the effect of chewing.

Suggest a more appropriate control experiment. Explain your suggestion.

[2 marks]

- 0 3 . 5 **Figure 2** shows the student's results.

Figure 2



Explain what these results suggest about the effect of chewing on the digestion of starch in wheat.

[3 marks]
